

# RYAN LEITE ALBUQUERQUE

SENIOR SOFTWARE ENGINEER

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Senior backend engineer with 26 years building production systems, currently the top contributor to a production ML platform for automated issue triage (FastAPI, Kafka, vector search on Kubernetes) while also delivering internal platforms end to end on his own – from data model to production deploy. Polyglot by necessity rather than preference, with production systems in Java, C, PHP, Ruby and Python, giving strong instincts for which problems are genuinely language-specific and which are architectural. Earlier career spent defining software configuration management practice at institute scale, including leading a 10-engineer team through a CMMI Level 3 appraisal as a trained SCAMPI evaluator. M.Sc. in Computer Science (Artificial Intelligence).

## EXPERIENCE

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### Senior Backend Software Engineer

Jan 2026 - present

CESAR (Centro de Estudos e Sistemas Avançados do Recife) • Recife, PE, Brazil

Backend and platform engineering for ML-driven developer-productivity systems, from APIs and data pipelines to Kubernetes delivery and AI/LLM tooling.

- Top 2026 contributor to a production ML platform for automated issue triage and duplicate detection: FastAPI, Kafka, Milvus vector search, Snowflake, Kubernetes on AWS EKS.
- Built solo a team onboarding platform (Django 6, GraphQL, Celery, Kubernetes) replacing a manual nine-step process.
- Built solo an engineering-metrics platform (Django + GraphQL, Nuxt 4, PostgreSQL on Kubernetes) with nine dashboards over coverage, LOC, pull-request and bug-fix data from CI.
- Added a self-service web app that triggers metric collection on demand with live log streaming, plus weekly Kubernetes CronJobs covering three repositories.
- Shipped AI developer tooling: an autonomous Claude Agent SDK bug-fixer that opens pull requests behind a confidence gate, and an MCP server serving six triage tools to Claude Code, Cursor and Xcode.

- Delivered two real-time team apps solo: planning poker on Django Channels WebSockets, and retrospectives on GraphQL with a six-state workflow engine.
  - Drove test-coverage and toolchain programs across five repositories: thousands of new tests, Poetry to uv, pre-commit to prek, ruff and type-check gates, CI coverage reporting.
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## Senior Backend Software Engineer

Oct 2022 - Dec 2025

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE,  
Brazil

Backend engineering and CI/CD ownership for two concurrent web platforms.

- Built and maintained backend services for two concurrent web platforms in Python (Django and Flask), enforcing production-grade quality through automated formatting, linting and test gates in CI.
  - Designed and implemented REST APIs to the JSON:API specification, standardizing response contracts across teams and reducing integration rework between backend and frontend squads.
  - Owned the CI/CD pipeline end to end (Docker, Git/GitFlow, Jenkins), delivering fully automated build, test and deployment cycles.
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## Senior Full-Stack Software Engineer

Jun 2013 - Sep 2022

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE,  
Brazil

Full-stack development, configuration management and technical leadership across four client projects over nine years.

- Delivered four production web applications using PHP (Yii), Python (Flask/Falcon), Ruby on Rails and JavaScript (Backbone/Marionette, EmberJS, VueJS).
  - Built and maintained CI/CD environments with Git/GitFlow, TeamCity, Jenkins, Ansible and HashiCorp Vault, enabling secure, repeatable and fully automated deployments.
  - Led technical direction for Scrum teams, running sprint planning, code review practice and delivery commitments.
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## Senior Full-Stack Software Engineer

Jan 2009 - Jun 2013

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE,  
Brazil

Full-stack and configuration management engineering across telephony, enterprise web and mobile projects.

- Developed a telephony platform in C on Asterisk, covering call flow logic and integration.
- Built an enterprise web and mobile application end to end using Java EE/ME, Android, Apache Struts 2, Apache CXF, Spring, Hibernate, Oracle and Apache Tomcat.
- Implemented a complete CI/CD environment (Subversion/Git, Maven, TeamCity, Tomcat) for automated builds and deployments.
- Built a Scrum task-board integration for Mantis (PHP) that gave distributed remote teams shared sprint visibility.
- Maintained a nationwide production web platform while shipping new features against live traffic.

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## Senior Software Configuration Management Engineer

Jan 2005 - Jan 2009

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE,  
Brazil

Leadership of the company-wide Software Configuration Management team and of the organization's CMMI 1.2 Level 3 appraisal.

- Led a 10-engineer Software Configuration Management team through a successful formal CMMI 1.2 Level 3 appraisal, serving as a trained SCAMPI evaluator.
  - Defined and maintained company-wide SCM processes as SCM lead within the Software Engineering Process Group (SEPG).
  - Rolled out SCM processes across multiple concurrent projects using Rational ClearCase, ClearDDTS, CVS, Subversion, Mantis, Apache Ant and Luntbuild.
  - Built automation tooling and Mantis extensions (PHP, Shell) for defect tracking, requirements traceability and code review reporting, later adopted as company-wide standards.
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## Software Configuration Management Engineer

Oct 2001 - Jan 2005

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE,  
Brazil

SCM process definition and rollout for multi-site mobile device development programs.

- Defined and implemented SCM processes for multi-site mobile device development projects using Rational ClearCase and ClearDDTS, contributing to the organization's CMM Level 2 certification.
  - Raised client evaluation scores for SCM performance by approximately 20% by automating configuration management activities across roughly twelve projects.
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## Junior Software Engineer

Jan 2001 - Oct 2001

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE,  
Brazil

Early-stage mobile application development on Java 2 Micro Edition (J2ME).

- Pioneered J2ME application development at CESAR, establishing the institute's first mobile development practice.
  - Developed 15+ mobile applications, primarily games, for early Java-enabled handsets.
  - Authored Sea Hunter, awarded Best Application from the Americas at the Asia Java Mobile Challenge.
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## Junior Software Engineer

Jun 2000 - Dec 2000

CESAR (Centro de Estudos e Sistemas Avançados do Recife) · Recife, PE,  
Brazil

Java development on a standard three-tier architecture.

- Developed a WAP application in J2SE for lottery game management.
- Built an e-commerce portal used to validate a team member's M.Sc. dissertation research.

SKILLS

BACKEND

Expert

Python, Django, FastAPI, Flask, Falcon, GraphQL, Strawberry GraphQL, REST API, JSON:API, Celery, Kafka, WebSockets, PHP, Yii, Laravel, Ruby on Rails, Java

FRONTEND

Advanced

JavaScript, TypeScript, VueJS, NuxtJS, PrimeVue, Tailwind CSS, Pinia, EmberJS, Backbone/Marionette, HTML, CSS, SCSS/SASS, BEM

AI & ML ENGINEERING

Advanced

LLM application development, Model Context Protocol (MCP), Claude Agent SDK, Agentic tooling, Vector search, Milvus, Embeddings, Sentence Transformers, PyTorch, Snowflake

DEVOPS & CI/CD

Expert

CI/CD, Jenkins, TeamCity, GitLab CI, Docker, Multi-arch builds, Ansible, HashiCorp Vault, Sealed Secrets, uv, mise, prek, ruff, pytest, Bash, Git, GitFlow, GitHub, GitLab

CLOUD & ORCHESTRATION

Advanced

Kubernetes, Helm, Kustomize, AWS EKS, AWS, Google Cloud, mTLS, OAuth 2.0, OpenTelemetry, Sentry

DATABASES

Advanced

PostgreSQL, MySQL, Oracle, Redis, Milvus, Snowflake

PROCESS & LEADERSHIP

Expert

Software Configuration Management, CMMI, SCAMPI appraisal, Scrum, Kanban, Technical Leadership, Code Review, Mentoring

## EDUCATION

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### Federal University of Pernambuco (UFPE)

Mar 2000 - Aug 2002

M.Sc. in Computer Science

Specialization in Artificial Intelligence. Dissertation: "KAGENT - A Java API for Intelligent Agents in Mobile Devices" (available at <https://shorturl.at/p0Zh0>).

Artificial Intelligence · Multi-Agent Systems · Software Engineering

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### Federal University of Ceará (UFC)

Jan 1996 - Jan 2000

B.Sc. in Computer Science

## PUBLICATIONS

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### **Software Configuration Management Diversity: A Commercial Case Study**

2008-07

BCS-CMSG - The CMDB and CMS: the Powerhouse of Service Management

Authors: T. H. B. Oliveira, R. Albuquerque.

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### **Desenvolvendo Projeto Conceitual de Jogos Wireless (Developing Conceptual Design for Wireless Games)**

2002-10

1st Brazilian Workshop on Games and Digital Entertainment (WJOGOS), SIBGRAPI 2002, Fortaleza, Brazil

Authors: D. R. Silva, A. Damasceno, E. Perazzo, P. Macedo, R. Albuquerque, G. L. Ramalho.

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### **Embedding a J2ME-Based Inference Engine in Handheld Devices: The KEOPS Case Study**

2002

Workshop on Ubiquitous Agents on Embedded, Wearable and Mobile Devices, Italy

Authors: R. Albuquerque, P. Guedes, C. Figueira Filho, J. Robin, G. Ramalho.

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## KSACI: A Handheld Device Infrastructure for Agent Communication

2001

ATAL 2001 Conference, Seattle, USA

Authors: R. Albuquerque, J. Hubner, G. Eliano de Paula, J. Sichman, G. Ramalho.

### AWARDS

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## Best Application from the Americas – Sea Hunter

2001

Asia Java Mobile Challenge

Awarded for Sea Hunter, a J2ME mobile game developed for early Java-enabled handsets.

### CERTIFICATES

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## AI for Developers (54h)

2025-08

CESAR School

## Certified ScrumMaster (CSM)

2017-10

Scrum Alliance

## Zend PHP Certified Developer

2014-10

Zend Technologies, Ltd.

### LANGUAGES

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#### PORTUGUESE

Native speaker

#### ENGLISH

Professional working proficiency